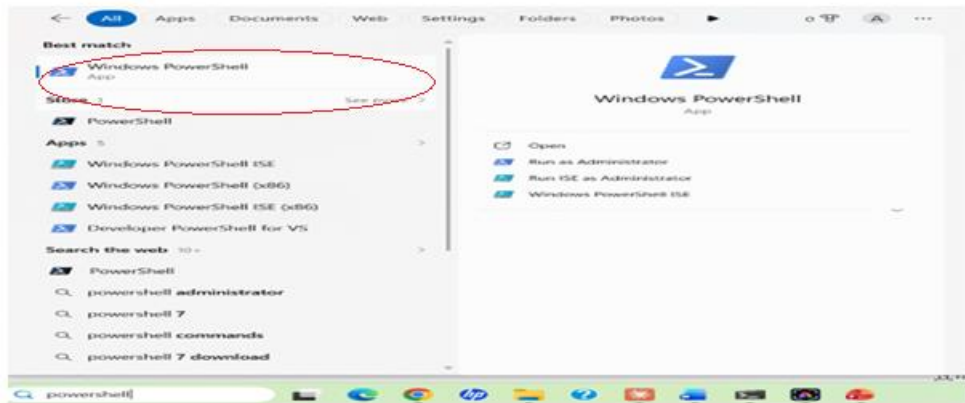
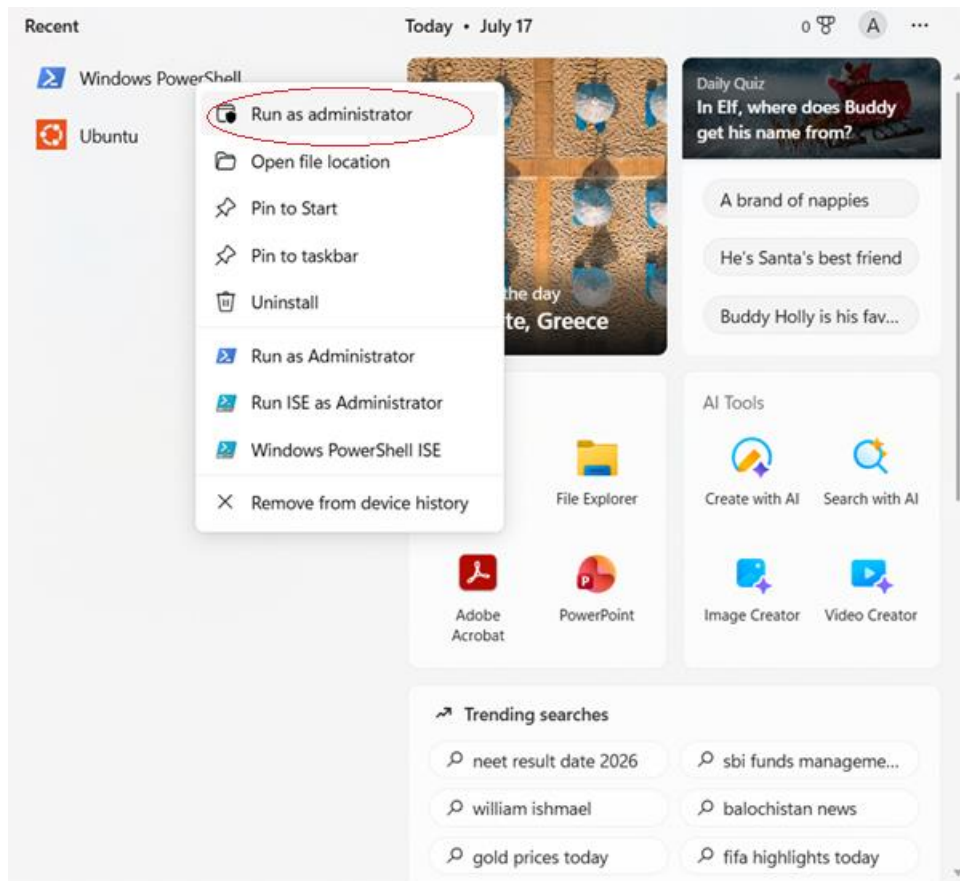


Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\windows\system32> wsl --install
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: arnav

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu

```
arnav@LAPTOP-BN8992KC:~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [293 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [357 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [76.5 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.1 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [91.1 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [138 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [53.1 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [153 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [43.9 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [255 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [245 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [47.6 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [45.5 kB]
Fetched 2149 kB in 7s (295 kB/s)
74 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Step 7: Install GCC (C Compiler)

```
arnav@LAPTOP-BN8992KC:~$ sudo apt install build-essential
Installing:
  build-essential

Installing dependencies:
  bzip2                gcc-x86-64-linux-gnu      libisl23
  cpp                  libalgorithm-diff-perl  libitm1
  cpp-15               libalgorithm-diff-xs-perl  liblsan0
  cpp-15-x86-64-linux-gnu libalgorithm-merge-perl  libmpc3
  cpp-x86-64-linux-gnu libasan8                  libquadmath0
  dpkg-dev             libc-dev-bin             libstdc++-15-dev
  fakeroot             libc6-dev                 libtsan2
  g++                  libcc1-0                  libubsan1
  g++-15               libcrypt-dev              linux-libc-dev
  g++-15-x86-64-linux-gnu libdpkg-perl               lto-disabled-list
  g++-x86-64-linux-gnu libfakeroot                make
  gcc                  libfile-fcntllock-perl   manpages-dev
  gcc-15               libgcc-15-dev             rpcsvc-proto
  gcc-15-base          libgomp1
  gcc-15-x86-64-linux-gnu libhwasan0

Suggested packages:
  bzip2-doc                gcc-15-doc      gcc-doc
  cpp-doc                  gcc-multilib    gcc-15-multilib
```

Step 8: Verify the Installation

```
arnav@LAPTOP-BN8992KC:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Step 9: Create Your C Program

Example: nano Processid.c

Step 10: Compile the Program

```

#include <stdio.h>
#include <unistd.h>

int main() {
    printf("Process ID: %d\n", getpid());
    printf("Parent Process ID: %d\n", getppid());

    return 0;
}

```

```

arnav@LAPTOP-BN8992KC:~/OS$ nano Processid.c
arnav@LAPTOP-BN8992KC:~/OS$ gcc Processid.c -o Processid
arnav@LAPTOP-BN8992KC:~/OS$ ./Processid
Process ID: 645
Parent Process ID: 320
arnav@LAPTOP-BN8992KC:~/OS$

```

GIT HUB